

FlowState calibration & validation report

MODEL INTEGRITY FAILURE — 3 of 20 run(s) locked (a permanent standstill); the no_locks acceptance criterion fails. Every metric of those runs includes the vehicles trapped behind the lock: see Model integrity and Limitations.

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Client summary

Baseline gate: NOT EVALUATED. No baseline gate result was supplied with this run set, so the model has not been shown to reproduce the corridor under the study protocol (docs/FRISCO_PROTOCOL.md section 6), and this report makes no strategy recommendation.

What we are confident about, and what we are not

Statement	Confident	Basis
Traffic counts at the detectors (link flows, C1)	not established	the baseline gate was not evaluated for this run set
Speeds at the detectors, 15-minute averages (C3)	not established	the baseline gate was not evaluated for this run set
Where and when the slowdowns form (bottlenecks, C6)	not established	the baseline gate was not evaluated for this run set
Stop-and-go wave speed (C4)	not established	the baseline gate was not evaluated for this run set
No simulated collisions (C5)	not established	the baseline gate was not evaluated for this run set
No permanent standstill (gridlock) in any run	no	3 of 20 run(s) locked, at on-ramp 178547099 (3), on-ramp 745524613 (2), off-ramp 18207390 (1), x 1350 m (1), on-ramp 1077665160 (1), on-ramp 769818012 (1), off-ramp 42165869 (1), on-ramp 53062592 (1), x 6550 m (1), x 5450 m (1), x 3250 m (1): a queue stood with no discharge for 10 min or more and its vehicles never finished (see Model integrity)
Fuel (model estimate, SUMO HBEFA emission class; not validated against measured fuel)	no	fuel is not validated: no measured fuel was compared, so every fuel figure is a model estimate (docs/FRISCO_PROTOCOL.md section 8.6)
Effects of the strategies	no	the baseline gate was not evaluated, so no strategy result is a finding
Robustness of strategy effects to driver and demand uncertainty	no	not evaluated in this report (docs/FRISCO_PROTOCOL.md section 8.5)
Waiting time on ramps and before entering the network	yes	counted: every run records total delay and travel time including waiting (docs/FRISCO_PROTOCOL.md section 8.2); no strategy recommendation is made because the run set has no single baseline (do-nothing) group to compare against

Strategy results

This report contains no strategy recommendations: the baseline gate was not evaluated. Any strategy table below is model output, not a finding.

Data, days and assumptions

- Calibration and validation days: not stated (no day split accompanies this report).
- Excluded detectors: not stated (no baseline gate result).
- Observed data: data/mndot/mndot_i94_wb_stpaul/observations.json, dates 20260901, 20260902, 20260903, 20260908, 20260909, 20260910, 20260915, 20260916, 20260917; mean over dates per

window (weekday typical profile).

- Ramp volumes estimated from mainline differences, and every split assumption behind them, are listed in the study's ramp-estimation artifact, not in this run set (docs/FRISCO_PROTOCOL.md section 2.3).
- Strategy assumptions: each strategy arm's penetration, compliance and settings are as its label states; they are assumptions, not measured behaviour.

Limits of this study

- Single corridor: the results describe this corridor, period and day set; transfer to other corridors is not established.
- Model-form uncertainty: the car-following and lane-change models' own assumptions are not captured by the seed-to-seed intervals.
- Every fuel figure is a model estimate (SUMO HBEFA emission class), not validated against measured fuel; differences in fuel between configurations are model predictions.
- Compliance and strategy assumptions: strategy results hold only for the settings simulated, under the configured driver population and demand.
- Results are reported as they came out, including failures.

Provenance

Profile: fhwa_tat3_2004. Seeds: 134183728835869882, 165503670820534583, 2378473973028931053, 3011106312394044631, 3747978530954135749, 3944094060050347669, 4910985839736976611, 5690692725577505498, 6134032994440706937, 6143473282319009404, 6538422657834023852, 661281422688282993, 677105600768189526, 6904272788004776631, 6914975401685141156, 6953598295321596746, 7382187975121682178, 8026499204807041784, 8557154790156791364, 887972120279483394.

Measurement window: each run's configured warm-up is discarded from every metric (warm-up per run, in seconds: 1800). Travel times keep whole journeys that begin inside the window and are measured over [1089.1, 11071.8] m. Fuel per vehicle-km, a model estimate (SUMO HBEFA emission class; not validated against measured fuel), remains a whole-run ratio unless the run records a post-warm-up fuel total.

Insertion: 678240 vehicles planned over 20 run(s), 648017 departed (0.955 of plan on average, lowest 0.889), 31026.2 arrived per run (over 20 of 20); verdict: backlog: 4 % of planned vehicles never departed.

Weave exits at on-ramp 745524613 (exit C-D split 18208090): 598 of 28565 reached exiters (2.09 %) were given up at the gore's end and rerouted through over 20 run(s), above the 2 % threshold; the exit's link flow is short by that count and every mainline link downstream carries it.

Weave exits at on-ramp 769818012 (exit off-ramp 18207598): 1017 of 77466 reached exiters (1.31 %) were given up at the gore's end and rerouted through over 20 run(s), within the 2 % threshold; the exit's link flow is short by that count and every mainline link downstream carries it.

Run	Config hash	Seed	Tier	seeded	Wall time [s]
134183728835869882	db9fbab5fc6e	134183728835869882	micro	seeded=False	897.3
165503670820534583	db9fbab5fc6e	165503670820534583	micro	seeded=False	775.6
2378473973028931053	db9fbab5fc6e	2378473973028931053	micro	seeded=False	920.7
3011106312394044631	db9fbab5fc6e	3011106312394044631	micro	seeded=False	1401
3747978530954135749	db9fbab5fc6e	3747978530954135749	micro	seeded=False	1725
3944094060050347669	db9fbab5fc6e	3944094060050347669	micro	seeded=False	1394
4910985839736976611	db9fbab5fc6e	4910985839736976611	micro	seeded=False	1162
5690692725577505498	db9fbab5fc6e	5690692725577505498	micro	seeded=False	1242
6134032994440706937	db9fbab5fc6e	6134032994440706937	micro	seeded=False	1308

Run	Config hash	Seed	Tier	seeded	Wall time [s]
6143473282319009404	db9fbab5fc6e	6143473282319009404	micro	seeded=False	945.2
6538422657834023852	db9fbab5fc6e	6538422657834023852	micro	seeded=False	1412
661281422688282993	db9fbab5fc6e	661281422688282993	micro	seeded=False	1003
677105600768189526	db9fbab5fc6e	677105600768189526	micro	seeded=False	1815
6904272788004776631	db9fbab5fc6e	6904272788004776631	micro	seeded=False	955.3
6914975401685141156	db9fbab5fc6e	6914975401685141156	micro	seeded=False	1308
6953598295321596746	db9fbab5fc6e	6953598295321596746	micro	seeded=False	1144
7382187975121682178	db9fbab5fc6e	7382187975121682178	micro	seeded=False	1018
8026499204807041784	db9fbab5fc6e	8026499204807041784	micro	seeded=False	1117
8557154790156791364	db9fbab5fc6e	8557154790156791364	micro	seeded=False	992.5
887972120279483394	db9fbab5fc6e	887972120279483394	micro	seeded=False	1044

Package versions (from run metadata)

- eclipse-sumo: 1.27.1
- flowstate_core: 2.0.0
- libsumo: 1.27.1
- microsim: 2.0.0
- numpy: 2.5.2
- pandas: 3.0.5
- pyarrow: 25.0.1
- python: 3.12.15

Calibration artifacts used

Artifact	data_hash
artifacts/idm_i24_capacity_amax_kl.0.json	aa97dd93d2bf250ea23c3624f91afe8d7035a672fd13efbd53013709e12c7d56

Observed data

The link-flow and segment-speed criteria are scored against this observed artifact: hourly volumes formed from the fully observed windows of each hour at every mainline station, and mean speeds per station segment (each station owns the span to the midpoints with its neighbours). Simulation time zero is the artifact's local start time; the run's warm-up, and any window the run does not cover to its end, are excluded, and a station-window the detector did not measure is skipped, never imputed — the coverage rows say how much of the grid was compared. A station whose cross-section lies outside the simulated position span is excluded from both comparisons rather than scored against a simulated flow of zero; the rows say how many were. When the artifact carries a detector-estimated backward wave speed, it is printed here as context for the band the simulated wave speed is scored against — it is a property of the corridor, not a target, and no criterion is evaluated from it.

Field	Value
artifact	data/mndot/mndot_i94_wb_stpaul/observations.json
corridor	I-94 WB
source provider	MnDOT RTMC Mayfly API
source dates	20260901, 20260902, 20260903, 20260908, 20260909, 20260910, 20260915, 20260916, 20260917
source url	https://data.dot.state.mn.us/mayfly
aggregation	mean over dates per window (weekday typical profile)
local clock time of simulation t = 0	05:30
observation window [s]	300

Field	Value
mainline stations compared	14
observation windows	48
windows inside the measurement window	42
station-windows with an observed flow [fraction]	1
station-windows with an observed speed [fraction]	1
link-hour comparisons (pooled over replicates)	840
speed cells compared (pooled over replicates)	11760
replicates scored against the observations	20
stations excluded (outside the simulated span)	0
detector-estimated backward wave speed (context, not a criterion)	median 21.3 km/h (IQR 18.6–24.2) from 6 of 13 station pairs; leave-one-date-out 18.4–21.6 km/h over 9 dates (fewest 5 pairs); the model's band is 14–22 km/h

Model integrity

What the runs' metadata records about the simulation itself misbehaving (docs/CONTRACTS.md). A collision in a car-following simulation is a model defect, not a traffic outcome, and so is a lock: vehicles standing with no discharge past a point while a queue builds behind it, read from each run's space-time table and vehicle table (validation.locks). A run whose metadata carries no counter, or that has neither table, is reported as not recorded, never as zero.

- Collisions: 0 over 20 run(s) that record the counter; per run 0 [0, 0] (two-sided 95 % t-interval, n = 20); 0 per 1,000 departed vehicles (0 over 648017 departed in 20 run(s), whole runs including warm-up).
- Locks: 3 of 20 run(s) locked (15 %; 95 % Clopper-Pearson interval 3.21–37.9 %). A lock is a queue standing with no discharge past a point for at least 10 min.
- Forced lane changes at on-ramp 18207436 (scripted merge): 861 of 9862 completed change(s) over 20 run(s) were made under SUMO's forced lane-change mode, in which SUMO refuses a change only on an overlap — the path around its own lane-change safety checks.
- Forced lane changes at on-ramp 178547099 (scripted merge): 2765 of 16315 completed change(s) over 20 run(s) were made under SUMO's forced lane-change mode, in which SUMO refuses a change only on an overlap — the path around its own lane-change safety checks.
- Forced lane changes at on-ramp 745524613 (weave section): 5710 of 16356 completed change(s) over 20 run(s) were made under SUMO's forced lane-change mode, in which SUMO refuses a change only on an overlap — the path around its own lane-change safety checks.
- Forced lane changes at on-ramp 769818012 (weave section): 9502 of 91052 completed change(s) over 20 run(s) were made under SUMO's forced lane-change mode, in which SUMO refuses a change only on an overlap — the path around its own lane-change safety checks.

Run	Head x [m]	Section	Onset [s]	Duration [min]	To the run's end	Trapped in network	Never departed upstream
301110631239404 4631	5316.6	on-ramp 745524613 (weave)	10110	≥ 71.5	yes	1928	3398
301110631239404 4631	4450.0	on-ramp 178547099 (merge)	10275	≥ 68.8	yes	1587	3108
301110631239404 4631	2350.0	off-ramp 18207390 (diverge)	10950	≥ 57.5	yes	867	1902
301110631239404 4631	1850.0	on-ramp 1077665160 (merge)	11475	≥ 48.8	yes	723	1902
301110631239404 4631	1350.0	x 1350 m	11460	≥ 49	yes	531	1902
677105600768189 526	10731.6	on-ramp 769818012 (weave)	11805	≥ 43.2	yes	3049	3294

Run	Head x [m]	Section	Onset [s]	Duration [min]	To the run's end	Trapped in network	Never departed upstream
677105600768189526	9250.0	off-ramp 42165869 (diverge)	12195	≥ 36.8	yes	2340	653
677105600768189526	7250.0	on-ramp 53062592 (merge)	12780	≥ 27	yes	1798	366
677105600768189526	6550.0	x 6550 m	12795	≥ 26.8	yes	1576	366
677105600768189526	5450.0	x 5450 m	12990	≥ 23.5	yes	1306	145
677105600768189526	4250.0	on-ramp 178547099 (merge)	13335	≥ 17.8	yes	873	31
8026499204807041784	5316.6	on-ramp 745524613 (weave)	12495	≥ 31.8	yes	1593	382
8026499204807041784	4250.0	on-ramp 178547099 (merge)	12780	≥ 27	yes	1154	140
8026499204807041784	3250.0	x 3250 m	13395	≥ 16.8	yes	598	0

Onset is when the head started standing (simulation time); $\leq$ marks an upper bound read at the run's end, $\geq$ a duration cut short by the run's end. Trapped: vehicles still in the network at the end, upstream of the head and bound past it. Never departed: vehicles of origins at or upstream of the head that never entered the network, any ordinary backlog of those origins included.

Acceptance criteria

Criterion	Value	Threshold	Evaluated	Result
link_flows_geh	0.5226	GEH < 5 for > 85% of link-hour comparisons	yes	FAIL – fraction of comparisons with GEH < 5
wave_speed	4.897	backward wave speed in [14, 22] km/h (emergent, unseeded)	yes	FAIL – detector: stack: slant-stack peak of the two-way demeaned field on 15 s x 75 m bins over front speeds [-40, -2] km/h in 0.25 km/h steps, peak/median contrast ≥ 3 , edge peaks rejected
ring_emergence	–	Sugiyama ring emergence benchmark reproduced	no	NOT EVALUATED – not evaluated: input not supplied
ring_dampening	–	Stern single-AV dampening benchmark reproduced	no	NOT EVALUATED – not evaluated: input not supplied
n_seeds	20	n_seeds ≥ 20	yes	PASS
sensitivity_grid	–	penetration {1%, 2%, 5%, 10%, 15%, 20%} x compliance {25%, 50%, 80%, 100%} grid published with CIs	no	NOT EVALUATED – not evaluated: input not supplied
no_collisions	0	zero SUMO collisions in every run, each run recording the counter	yes	PASS – no collision in 20 run(s); FlowState internal standard (CLAUDE.md §3.3; owner decision 2026-10-04), not an FHWA or DOT criterion
no_locks	3	no run locked (no queue standing without discharge for 10 min or more), every run recorded	yes	FAIL – 3 of 20 recorded run(s) locked; FlowState internal standard (CLAUDE.md §0.1; docs/I94_COLLAPSE_DIAGNOSIS.md §7, 2026-10-07), not an FHWA or DOT criterion

A row marked not evaluated is never a pass: either no input was supplied, or the value was measured with a recipe this profile does not accept (the row's result says which). An unevaluated criterion counts as failing. A row marked not recorded is never a pass either: some run in the set carries no record of what the row checks, and a missing record is not a zero. The no_collisions row is a FlowState model-integrity requirement, not an FHWA criterion: every run must record its SUMO collision counter, and the total must be zero. So is the no_locks row: every run must carry a lock record, and no run may

lock.

Wave-speed criterion input: mean over 19 of 20 unseeded replicate(s) of group merge scripted on on-ramp 18207436 + merge scripted on on-ramp 178547099 + merge weave on on-ramp 745524613 + merge weave on on-ramp 769818012 (db9fbab5fc6e), measured with the stack detector on its own bins; 1 replicate(s) detected no backward front and are excluded from the mean; fewer than 20 contributing replicate(s) — underpowered, not a headline value.

Speed criterion by time aggregation

The segment-speed criterion compares a replicate mean with one recorded day. The floor column is the recorded field against its own three-window moving average: below that resolution the day does not repeat itself, so no ensemble mean can score under the floor. The criterion row keeps its native resolution; this table says how much of its value is resolution.

Aggregation	RMSPE, replicate mean vs observed	Floor: observed vs its own three-window average
5 min (criterion)	0.4113	0.02832
15 min	0.4071	0.06799
30 min	0.3977	
60 min	0.3909	
whole period	0.3426	

Metrics

Mean with two-sided t-distribution confidence bounds at the 95 percent level over n replicates. Rows flagged underpowered have fewer than 20 replicates and must not be quoted as headline results. Travel-time rows rest on the vehicles counted by the `n_travel_time_veh` row, which are those that entered within the measurement window and completed the span.

The `wave_speed_kmh` row is the metrics detector's diagnostic reading (standard); the acceptance criterion above is measured separately with the profile's stack detector on its own bins, so the two values can differ.

Rows marked (model estimate) carry fuel (model estimate, SUMO HBEFA emission class; not validated against measured fuel). SUMO computes fuel from its HBEFA emission classes; this study has measured no fuel to check it against (docs/FRISCO_PROTOCOL.md section 8.6).

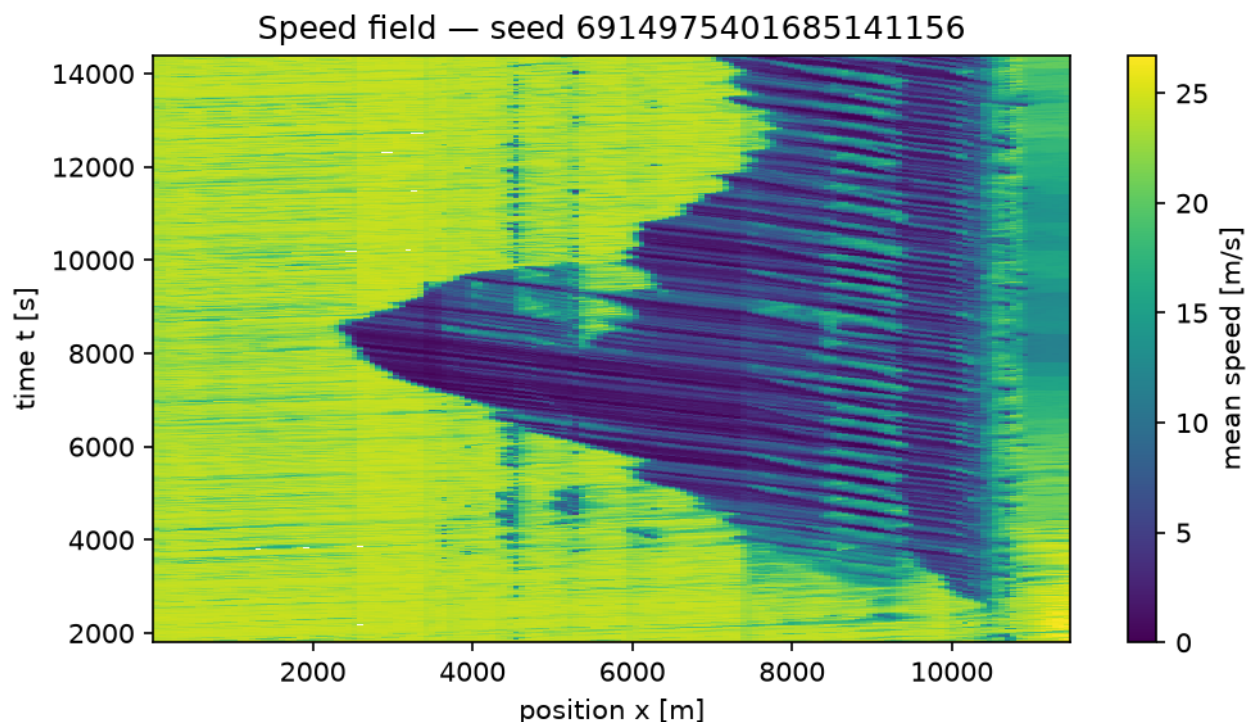
merge scripted on on-ramp 18207436 + merge scripted on on-ramp 178547099 + merge weave on on-ramp 745524613 + merge weave on on-ramp 769818012 (db9fbab5fc6e)

Replicates: n = 20 distinct seeds (134183728835869882, 165503670820534583, 2378473973028931053, 3011106312394044631, 3747978530954135749, 3944094060050347669, 4910985839736976611, 5690692725577505498, 6134032994440706937, 6143473282319009404, 6538422657834023852, 661281422688282993, 677105600768189526, 6904272788004776631, 6914975401685141156, 6953598295321596746, 7382187975121682178, 8026499204807041784, 8557154790156791364, 887972120279483394); replicate criterion (`n_seeds` >= 20): PASS.

Metric	Mean	Lower	Upper	n	Underpowered
throughput_veh_h	2866	2755	2976	20	no
mean_tt_s	572.6	547.7	597.5	20	no
p90_tt_s	1352	1279	1425	20	no
sigma_v_spatial_ms	7.669	7.474	7.864	20	no
sigma_v_temporal_ms	4.897	4.824	4.971	20	no
vmt_veh_km	1.288e+05	1.253e+05	1.322e+05	20	no
vht_veh_h	4231	3957	4506	20	no

Metric	Mean	Lower	Upper	n	Underpowered
fuel_ml_per_veh_km (model estimate)	112.5	105.4	119.7	20	no
wave_count	35.25	31.88	38.62	20	no
wave_speed_kmh	7.926	7.627	8.225	20	no
wave_amplitude_ms	15.36	15.22	15.51	20	no
n_travel_time_veh	1.236e+04	1.205e+04	1.267e+04	20	no

Speed contours



Space-time mean-speed contour, seed 6914975401685141156

Limitations

- This run set contains 3 locked run(s) of 20 (3011106312394044631 at on-ramp 745524613 from 10110 s, on-ramp 178547099 from 10275 s, off-ramp 18207390 from 10950 s, on-ramp 1077665160 from 11475 s, x 1350 m from 11460 s; 677105600768189526 at on-ramp 769818012 from 11805 s, off-ramp 42165869 from 12195 s, on-ramp 53062592 from 12780 s, x 6550 m from 12795 s, x 5450 m from 12990 s, on-ramp 178547099 from 13335 s; 8026499204807041784 at on-ramp 745524613 from 12495 s, on-ramp 178547099 from 12780 s, x 3250 m from 13395 s). A lock is a model defect, not a traffic outcome: the vehicles trapped behind it never finish, so the travel times of those runs are censored and their flows past the lock fall to zero, and every metric and interval above includes them; see Model integrity.
- Results cover a single corridor/scenario family; transfer to other corridors is not established.
- Model-form uncertainty (IDM/SUMO car-following assumptions, effective single-pipe demand representation) is not captured by seed-to-seed confidence intervals.
- AV penetration and compliance are swept assumptions, not measured behavior; conclusions hold only within the swept grid.
- Metrics from fewer than 20 replicates are flagged underpowered and are diagnostics, not claims.
- Every fuel figure is a model estimate (SUMO HBEFA emission class), not validated against measured fuel; differences in fuel between configurations are model predictions.
- Measurement window: each run's configured warm-up is discarded from every metric (warm-up per run, in seconds: 1800). Travel times keep whole journeys that begin inside the window and are

measured over [1089.1, 11071.8] m. Fuel per vehicle-km, a model estimate (SUMO HBEFA emission class; not validated against measured fuel), remains a whole-run ratio unless the run records a post-warm-up fuel total.

- The observed comparison covers the artifact's stations, windows and days only; station-windows the detectors did not measure are excluded from both criteria rather than filled in, so the coverage rows bound how much of the corridor and period the two rows actually speak for.